THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

Dec 18, 2015 TIME: 9:30 am – 12:30 pm

I. Answer ALL questions. Section A carries 30 marks, and Section B carries 70 marks.

II. Atomic weights: H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K 39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

III. Properties of water:

Density = 1000 kg/m^3
Viscosity (15 °C) = 1.1404 mPa·s
Specific heat = 4.18 kJ/kg.°C
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion = 333 kJ/kg

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (10 multiple choice questions with 3 marks each)

1. Which of the following air pollutants is not monitored under Hong Kong's Air Quality Health Index? (a) Ground-level ozone, (b) sulphur dioxide, (c) $PM_{2.5}$, (d) CO, or (e) none of the above.
2. The alkalinity (ALK) of a water solution can be increased by the addition of chemicals. Which of the following chemicals is not able to increase ALK? (a) Na_2CO_3, (b) NaOH, (c) KOH, (d) $Al_2(SO_4)_3$, or (e) none of the above.
3. A water solution containing 1 mM Ca^{2+} and 2 mM F^- is in contact with the mineral $CaF_2(s)$. It is known that $CaF_2(s)$ has a solubility constant pK_{sp} of 10.5. Which of the following statements is the most appropriate? (a) The solution is in equilibrium with the $CaF_2(s)$, (b) the solution is under-saturated, (c) $CaF_2(s)$ precipitation is expected to happen, (d) $[Ca^{2+}]$ is expected to increase over time, or (e) none of the above.
4. Which of the following water quality parameters is not directly related to the organic strength of wastewater? (a) BOD, (b) TOC, (c) COD, (d) DO, (e) theoretical oxygen demand, or (f) none of the above.
5. Which of the following statements is correct for a typical eutrophic lake? A eutrophic lake has (a) low turbidity, (b) low nutrient level, (c) low diversity, (d) low suspended solids content, (e) low chlorophyll content, or (f) none of the above.
6. Which biological treatment belongs to anaerobic treatment? (a) Activated sludge, (b) aerated lagoon, (c) trickling filter, (d) rotating biological contactor, (e) membrane bioreactor, or (f) none of the above.
7. Which of the following treatment processes is not designed for particle removal? (a) Sedimentation, (b) coagulation, (c) flocculation, (d) filtration, (e) microfiltration, or (f) air stripping.
8. Which compound is considered as a greenhouse gas? (a) CO, (b) NO_2, (c) N_2, (d) O_2, (e) N_2O, or (f) none of the above.
9. A wastewater sample is filtered by a 0.45-μm glass filter. The solids retained by the filter are heated in an oven at 500 °C. The remaining fraction of the solids after heating is known as (a) TDS, (b) TSS, (c) TVS, (d) TFS, (e) none of the above.
10. Estimate the oxygen concentration in ambient air: (a) 1,400,000 g/m^3, (b) 300,000 g/m^3, (c) 1,400 g/m^3, (d) 300 g/m^3, (e) 1.4 g/m^3, (f) 0.3 g/m^3.

Section B (Three questions with unequal marks)

11. Thermal stratification can greatly affect the water quality of lakes.

    (a) Briefly explain the phenomenon of thermal stratification. Also sketch the temperature profile in a typical stratified lake. When is thermal stratification more likely to occur?
    (10 marks)

    (b) For a stratified eutrophic lake, reduced chemical species such as hydrogen sulfide (H_2S) and methane (CH_4) are commonly found in the hypolimnion (i.e., the bottom of a stratified lake). Briefly explain the reason(s) behind this phenomenon.
    (6 marks)

    (c) Briefly discuss the environmental significance of hydrogen sulfide.
    (4 marks)

12. (a) The atmospheric temperature versus altitude data are given in the table below. Determine the mixing depth. Use dry adiabatic lapse rate in your calculation.

Altitude (m)	0	100	200	300	400	500	600	700	800
Temperature (°C)	30	28	26	24	24	24	24	24	24

(6 marks)

(b) A wastewater of 1000 m^3/d is treated in a sewage lagoon. The concentration of biodegradable organic matter is 150 mg/L, and that in the treated effluent shall not exceed 20 mg/L. It is assumed that the biodegradable organic matter is removed in the sewage lagoon with a first order reaction rate of 0.3 day^{-1}. Perform a simple mass balance analysis and determine the minimum volume required for the sewage lagoon. List the important assumption(s) you made in your analysis.

(14 marks)

13. (a) A wastewater has a BOD_5 of 300 mg/L at 25°C. The ultimate BOD is 400 mg/L. Determine the rate constant at 25°C. Also estimate BOD_5 at 15°C. You can assume a temperature correction factor of 1.047 in your calculation(s). Based on your calculation(s), do you think that biological wastewater treatment is more effective in colder weather?

(10 marks)

(b) Municipal wastewater is commonly treated by conventional activated sludge (CAS) treatment. Sketch a CAS process diagram. Briefly explain the purpose of each unit treatment process.

(8 marks)

(c) F/M ratio is an important parameter in CAS. Explain the term F/M ratio.

(5 marks)

(d) Membrane bioreactor (MBR) technology is becoming increasingly popular. It has a good potential to replace CAS for wastewater treatment. What is MBR? What are the advantages and limitations of MBR?

(7 marks)

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